Online Appendix to
"What Survives a Placebo? Entropy, Market Coupling and the Plausibility of Stress Scenarios"

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This appendix collects material that supports the main text but is not needed to follow it: the dynamic postulates and the volume-time argument behind the framework (section OA.1); the full placebo tables (section OA.2); the severity ladder, the composition of scenarios and reverse stress testing (section OA.3); the computation of the EBA scenario's cost (section OA.4); and a euro-area application of the block decomposition, the entropic CISS (section OA.5). Numbers are those of the replication package. Table and figure numbers refer to this appendix unless they are prefixed "main text".

OA.1 Dynamic postulates and the volume-time argument

OA.1.1 The bridge from order flow to Gaussian prices

Over an interval, let V be traded volume and Q net signed order flow (buys minus sells). Two assumptions suffice. *Linear impact*: the change in log price is λQ, as in Kyle (1985). *Variance proportional to volume*: given V, the market knows only the mean of Q and a variance $\varsigma^2 V$. Maximum entropy then gives

$$\Delta X \mid V \;\sim\; \mathcal{N}\Big(\lambda m,\; \lambda^2 \varsigma^2 V\Big)\,. \tag{1}$$

The conditioning variable is volume, not time: this is the mixture-of-distributions hypothesis of Clark (1973) and Tauchen and Pitts (1983), and Ané and Geman (2000) show that returns measured in transaction time are close to Gaussian. Unconditionally, the return is a mixture of Gaussians, hence fat-tailed (Mandelbrot and Taylor, 1967): fat tails alone do not prove that information is present. If increments are moreover independent in volume time, the log price is a time-changed Brownian motion, and the price a geometric Brownian motion. Both assumptions are imperfect (real impact is closer to a square root, and order flow is autocorrelated), which is why all empirical work is done in calendar time.

OA.1.2 Index dynamics: a theorem and two postulates

The theorem. For a return X and a signal Y, $\mathbb{E}_Y[h(X \mid Y)] = h(X) - I(X;Y) \le h(X)$ (Cover and Thomas, 2006). Receiving information lowers entropy *on average* by exactly the mutual information. A particular realisation can raise it (a confusing signal); the jump component is therefore mainly, not exclusively, one-sided.

Postulate 1 (relaxation). Without new information, $\mathcal{J}$ reverts to a long-run level: $d\mathcal{J}_t = -\kappa(\mathcal{J}_t - \bar{\mathcal{J}})\,dt + \sigma_{\mathcal{J}}\,dW_t$. An Ornstein–Uhlenbeck process is the only process that is Markovian,

Gaussian and stationary (Doob, 1942). Adding the jumps caused by news gives

$$d\mathcal{J}_t \;=\; -\kappa(\mathcal{J}_t - \bar{\mathcal{J}})\,dt \;+\; \sigma_{\mathcal{J}}\,dW_t \;+\; dZ_t, \tag{2}$$

where Z_t is the sum of positive jumps up to t: the class of Barndorff-Nielsen and Shephard (2001). The half-life of a shock is $\ln 2/\kappa$.

Postulate 2 (sign). $\mathcal{J}$ is an even function of skewness: a distribution with a long left tail (a crash) and its mirror image (a rally) have the same $\mathcal{J}$. Saying anything about direction requires an outside ingredient. We postulate that jumps come with negative skewness, relying on the leverage effect (Black, 1976; Engle and Ng, 1993).

Why dependence and tails should rise together. Major news hits every asset on the same day. On that day, assets move together (coupling rises) and each records an unusual loss (its tails fatten). The two channels describe the same day seen from two places. In calm markets, large moves are asset-specific and couple nothing. This is H6. The null shows, however, that this coincidence is also produced mechanically by estimation.

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$$d\mathcal{J}_t \;=\; -\kappa(\mathcal{J}_t - \bar{\mathcal{J}})\,dt \;+\; \sigma_{\mathcal{J}}\,dW_t \;+\; dZ_t, \tag{3}$$

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OA.2 Full placebo tables and estimation details

OA.2.1 Empirical details

Estimators. Two covariance estimators run in parallel: a 252-day rolling window with the Ledoit and Wolf (2004) correction, and exponential weighting with a 60-day half-life and 10 % shrinkage towards the diagonal. The second prevents an extreme day *leaving* the window from creating a false jump. The shape deficit uses the Hyvärinen (1998) approximation, whose two terms separate a skewness channel and a tail channel; the better known Edgeworth expansion $S^2/12 + K^2/48$ overstates the deficit of a Student t with 4 degrees of freedom by a factor of 69. Jumps are detected at four times a local scale given by bipower variation lagged by one day (Barndorff-Nielsen and Shephard, 2004).

Inference. Two consecutive rolling windows share 251 of their 252 days. Regime comparisons therefore use a circular block bootstrap with 252-day blocks. The relaxation p-value follows the Dickey–Fuller distribution. Forecasts (H7) are re-estimated every year on the past and compared on dates spaced by the forecast horizon, with the Diebold and Mariano (1995) test.

Raw stress-regime results (S&P 500). Under stress (1,076 dates against 6,481), scale rises by 5.97 nats, dependence falls by 0.99, shape rises by 0.27 ($p = 0.16$), and $\mathcal{J}$ rises by 1.94 ($p < 0.001$, but without passing the null). Average correlation goes from 0.26 to 0.32, the effective number of modes from 13.1 to 11.4. Market skewness does not change significantly between regimes: the skewness signature is a property of jumps, not of the regime. The index shares about a fifth of its variance with volatility ($r = 0.46$), and the shape channel is orthogonal to it ($r = 0.08$).

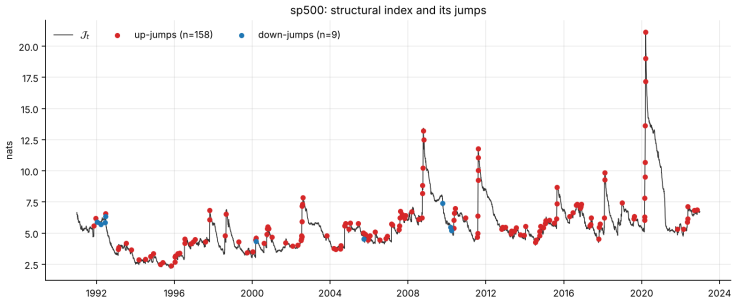


Figure OA.1: The structural index $\mathcal{J}_t$ on twenty S&P 500 stocks, 1990–2022, with detected jumps (red: rises, blue: falls). The sawtooth is visible, but the i.i.d. null reproduces it.

			i.i.d. null		CCC-FHS null		
	Statistic	Obs.	mean	p	mean	p	Verdict
H1	gap H^{dep}	-0.99	-0.22	< 0.001	0.06	< 0.001	**passes both**
H1	gap $\mathcal{J}$	1.93	2.13	0.698	0.81	0.003	CCC-FHS only
H1	gap $\mathcal{D}$	0.27	1.24	1.000	0.30	0.554	fails
H2	corr($\Delta H^{vol}, \Delta H^{dep}$)	0.10	0.23	0.298	0.55	0.031	CCC-FHS only
H2	β (stress)	0.20	0.06	0.415	-0.20	0.012	CCC-FHS only
H3	positive jumps	158.0	148.0	0.136	163.3	0.728	fails
H3	negative jumps	9.0	3.1	1.000	3.1	0.994	wrong side
H3	mean positive jump (nats)	0.62	1.13	1.000	0.47	0.002	CCC-FHS only
H3	skewness of $\Delta\mathcal{J}$	12.26	14.83	0.925	9.39	0.061	fails
H3	var. share of pos. jumps	0.838	0.900	0.987	0.752	< 0.001	CCC-FHS only
H4	half-life (days)	120.8	91.3	0.002	50.5	< 0.001	**passes both**
H4	sd($\mathcal{J}$)	2.03	1.54	0.007	0.60	< 0.001	**passes both**
H4	range($\mathcal{J}$)	18.95	9.99	< 0.001	4.75	< 0.001	**passes both**
H5	gap ΔS_{mkt}	-0.053	-0.010	0.221	-0.093	0.959	fails
H5	corr($\Delta\mathcal{J}, \Delta S_{mkt}$)	-0.243	-0.035	0.107	-0.381	0.948	fails
H6	coupling, stress	0.63	0.90	1.000	0.35	0.035	CCC-FHS only
H6	coupling gap	0.590	0.764	0.992	0.417	0.140	fails
–	corr($\mathcal{J}$, log vol)	0.461	0.412	—	0.519	—	—

Table OA.1: S&P 500: every placebo statistic against the two nulls, 999 replications each. p: exact rank p-value $(1 + k)/(1 + B)$ in the predicted direction. "Wrong side": both nulls exceed the data. Gaps: stress minus calm. Episode statistics and the verdicts with multiple-testing control are in the paper's table 5.

			i.i.d. null		CCC-FHS null		
	Statistic	Obs.	mean	p	mean	p	Verdict
H1	gap H^{dep}	-3.65	-0.58	< 0.001	-0.67	< 0.001	**passes both**
H1	gap $\mathcal{J}$	5.93	8.04	0.971	3.54	0.016	CCC-FHS only
H1	gap $\mathcal{D}$	0.98	4.86	1.000	1.04	0.461	fails
H2	corr($\Delta H^{vol}, \Delta H^{dep}$)	-0.63	-0.36	0.004	-0.34	0.005	**passes both**
H2	β (stress)	0.23	0.09	0.055	0.12	0.396	fails
H3	positive jumps	182.0	182.2	0.527	193.4	0.888	fails
H3	negative jumps	3.0	7.6	0.046	3.7	0.537	i.i.d. only
H3	mean positive jump (nats)	1.70	3.47	1.000	1.30	0.004	CCC-FHS only
H3	skewness of $\Delta\mathcal{J}$	17.04	18.32	0.677	14.35	0.201	fails
H3	var. share of pos. jumps	0.889	0.943	0.975	0.823	0.013	CCC-FHS only
H4	half-life (days)	189.8	100.6	< 0.001	52.8	< 0.001	**passes both**
H4	sd($\mathcal{J}$)	6.37	5.49	0.092	2.00	< 0.001	CCC-FHS only
H4	range($\mathcal{J}$)	54.47	37.28	0.004	18.53	< 0.001	**passes both**
H5	gap ΔS_{mkt}	-0.112	-0.059	0.119	-0.107	0.398	fails
H5	corr($\Delta\mathcal{J}, \Delta S_{mkt}$)	-0.440	-0.205	0.076	-0.520	0.824	fails
H6	coupling, stress	0.76	0.94	1.000	0.71	0.219	fails
H6	coupling gap	0.411	0.636	1.000	0.533	0.843	fails
–	corr($\mathcal{J}$, log vol)	0.677	0.442	—	0.670	—	—

Table OA.2: FF49 industries, 1990–2026: same conventions as table OA.1.

OA.3 Severity ladder, composition and reverse stress testing

OA.3.0.1 Calibrating the radius on the market itself

It remains to choose η. We read it from the market's information flow, that is, from how much the market itself revises its distribution over 21 days, $I_t^{(21)} = D_{\mathrm{KL}}(p_t \,\|\, p_{t-21})$, measured on the same quantity as the constraint (the portfolio distribution). Historical quantiles of this flow give a radius for each return period (table OA.3).

Return period	1 yr	2 yrs	5 yrs	10 yrs	20 yrs	50 yrs†
radius η (nats)	0.017	0.045	0.149	0.408	0.634	1.369
directional shock (std. dev.)	−0.08	−0.13	−0.24	−0.38	−0.47	−0.66
volatility multiplier	1.12	1.20	1.37	1.63	1.80	2.21
one-day ES_{99}	3.9%	4.2%	4.9%	5.9%	6.6%	8.3%
historical: realised volatility increase	1.54	1.77	2.28	3.25	3.57	4.43

Table OA.3: Severity ladder for an equally weighted portfolio of the twenty stocks, calibrated on the 21-day information flow (383 non-overlapping observations). Current state: annual volatility 20 %, one-day ES_{99} 3.4 %. Last row: quantile, at the same return period, of the ratio of next-21-day volatility to past-252-day volatility. † Extrapolated beyond 33 years of data.

The scenario spends its budget mostly on volatility rather than direction: at the ten-year rung, 63 % more volatility and a 0.38-standard-deviation shift, because ES is more sensitive to scale than to location. The radius depends on the portfolio: a ladder must be calibrated for the portfolio it is applied to.

A serious limit. The last row confronts the ladder with history. A ten-year scenario multiplies volatility by 1.6, whereas the ten-year quantile of the realised volatility increase is 3.2. The ladder is therefore *too mild* by a factor close to two. Either the information flow, measured on smoothed estimators, understates the true revision, or a Gaussian ball is the wrong geometry for such abrupt volatility shocks. We do not settle between the two. The return periods in the table must therefore not be quoted as frequencies. What withstands this limit are the *relative* comparisons of the next two subsections.

OA.3.0.2 Composing a scenario: volatility or correlation?

The entropy ball fixes the severity *of the portfolio* (its stressed mean and volatility). It does not say how each asset's covariance should move to get there. The default choice, used so far, is a uniform dilation of volatilities with correlations unchanged. That is a convention, not a fact about crises. Yet section 5 of the main text shows that, under stress, dependence moves on its own.

A simple property allows for this without changing severity. Portfolio variance, $w^\top \Sigma w$, is linear in Σ. If two covariances give the same portfolio variance, so does any mixture of the two (proposition in Appendix A of the main text). One can therefore mix an "all-volatility" covariance

and an "all-correlation" covariance with a weight α, the *dependence share*, without touching severity. One caveat: correlation alone cannot carry any severity. With all correlations equal to one, our portfolio's volatility cannot exceed 1.5 times its current level; beyond that, the rest comes from dilation.

The dependence share is calibrated on the robust results. The *anchor* comes from H1: under stress, dependence accounts for $0.99/(0.99 + 5.97) = 14\,\%$ of the move in Gaussian entropy (scale plus dependence). The *refinement* comes from H8: over the more severe half of episodes, this share reaches 49 %.

Composition of the ten-year scenario ($\eta = 0.41$)	dependence share	total cost (nats)
default: uniform dilation	0.00	6.80
calibrated on H1 (regime average)	0.14	2.88
refined by H8 (severe episodes)	0.49	0.71

Table OA.4: Total cost in nats, on the twenty assets, of three compositions that produce exactly the same portfolio severity ($\mathrm{ES}_{99} = 5.9\,\%$). On the French industries: 14.7, 8.3 and 5.8 nats for shares of 0, 0.20 and 0.75.

The result is counter-intuitive (table OA.4): the larger the dependence share, the *less* improbable the scenario, for the same severity. Uniform dilation is therefore not a prudent choice; it is both less realistic and more costly in plausibility than a composition anchored on history. This finding is specific to this panel and should not be read as a general law.

OA.3.0.3 Reverse stress testing

The same unit also answers the reverse question: what is the least improbable scenario that produces a given loss? On the main panel, the daily loss exceeded one day in a hundred (3.2 %) costs 1.5 nats; the one exceeded one day in a thousand (7.7 %) costs 4.8 nats; the worst day in 33 years (11.5 %) costs 9.0 nats. Beyond a few nats, the non-parametric version of this calculation rests on a handful of days, and the Gaussian version should be preferred.

OA.4 Computing the cost in nats of the EBA 2025 adverse scenario

Code: scripts/eba_nats.py; results: paper/tables/eba_nats.csv.

1. The shocks. We take the adverse macro-financial scenario (European Banking Authority, 2025) for 2025, as deviations from the baseline (annual averages): EU equities $-50\,\%$, the baseline assuming them unchanged (table 4.1.7 of the document); German long rate 3.50 % against 2.40 % and EU long rate 4.80 % against 3.12 % (table 4.1.6); euro exchange rates against the dollar, pound, yen and Swiss franc unchanged (table 4.1.10). Each shock is mapped to the corresponding series of the euro-area panel (section OA.5), giving an eleven-component vector Δ:

Cost in nats	equities and rates	with FX fixed
EBA 2025 adverse scenario, year 1	**10.5**	**11.1**
equities only / rates only	6.8 / 3.1	
sum of variable-by-variable costs	31.9	
Observed years: 2011 / 2008 / 2022	3.1 / 11.2 / 11.9	3.1 / 18.2 / 13.2
worst rolling year (March 2008–March 2009)	14.4	19.5

Table OA.5: Cost in nats of the EBA 2025 adverse scenario and of observed years, on five equity indices, two ten-year rates and four exchange rates, with the annual covariance estimated on 2004–2024.

EBA shock	Panel series	Δ_i	σ_i (1 yr)	cost alone (nats)
EU equities −50 %	DAX, CAC 40, FTSE MIB, IBEX 35, AEX	$\ln 0.5 = -0.693$	19–23 %	4.4–6.5
German long rate	10-year AAA yield (ECB)	+1.10 pt	0.65 pt	1.4
EU long rate	10-year all-ratings yield (ECB)	+1.68 pt	0.68 pt	3.0
FX unchanged	EUR/USD, GBP, JPY, CHF	0	7–11 %	0

Equities as log returns, rates as level changes; the maturity of long rates is not specified by the document and is taken at ten years. GDP, unemployment, real estate and inflation, which are not market prices, are not used.

2. The reference. $p_0 = \mathcal{N}(0, \Sigma)$, where Σ is the covariance of one-year changes in the eleven series, estimated on daily changes from 6 September 2004 to 31 December 2024 (the data available to the designer) and multiplied by 252.

3. The formula. The scenario fixes means, not volatilities or correlations. We read it as a pure mean shift, hence (Appendix A of the main text) $D_{\mathrm{KL}}(p_{\mathrm{scenario}} \,\|\, p_0) = \frac{1}{2}\Delta^\top \Sigma^{-1} \Delta$. This is the *minimal* cost consistent with the scenario: among all distributions with mean Δ, the shifted Gaussian is the closest to p_0. Unspecified variables (two- and five-year rates, banks, etc.) are left free; filling them with their conditional expectation is the cheapest choice, and the cost then depends only on the shocked block S: $\frac{1}{2}\Delta_S^\top \Sigma_{SS}^{-1} \Delta_S$. The cost of a single variable is $\frac{1}{2}(\Delta_i/\sigma_i)^2$; for the DAX ($\sigma = 20.3\,\%$), the shock is 3.4 standard deviations and costs 5.8 nats. Read η nats as $\sqrt{2\eta}$ standard deviations.

4. Reading. The sum of variable-by-variable costs is 31.9 nats, the joint cost 10.5: a common shock to highly correlated indices follows the most likely direction in the data. Equities alone (6.8) plus rates alone (3.1) leave 0.5 nat of interaction (falling equities with rising rates go against the historical correlation, +0.13). Fixing FX adds 0.7 nat.

5. Observed years. Same calculation, same Σ, on the realised changes of the eleven series (31 December to 31 December; worst 252-day window ending 17 March 2009). The median year

costs 4.1 nats. The scenario combines the equity fall of 2008 (when rates fell) and the rate rise of 2022 (close to 3 points).

6. Variants (equities and rates / with FX fixed). Covariance of rolling annual changes: 12.8 / 18.7; sample to 2026: 10.6 / 11.3; crisis covariance 2008–2012: 8.4 / 9.0. At the end of the three-year horizon (2027: −42 %, +87 and +112 bp, three-year covariance), the cost falls to 2.1 / 2.2: severity is concentrated in the first year.

7. Limits. Shocks in annual averages; the EU equity fall applied to each index; a Gaussian reference without fat tails, which tends to overstate the cost. The ECB's instantaneous market-risk scenario is not priced (detailed tables not accessible), but the same formula applies. These costs (eleven series, one year) are not directly comparable with table OA.3 (equity portfolio, 21 days).

OA.5 The entropic CISS: a euro-area application of the block decomposition

The paper's one robust result, coupling under stress, becomes here an indicator in the format of the ECB's CISS (Holló et al., 2012). Like the CISS, it covers the money market, bond, equity, financial intermediary and foreign exchange segments. Unlike the CISS, it says which share of coupling lies within each segment and which lies *between* segments. For a scenario designer, this is precisely the information needed to set correlations between shocks to different markets.

OA.5.1 An exact decomposition by segment

Split the series into segments $B_1, \ldots, B_K$. Total coupling decomposes exactly:

$$C(R) \;=\; \underbrace{\sum_{k=1}^{K} C(R_{kk})}_{\text{within segments}} \;+\; \underbrace{I(B_1;\ldots;B_K)}_{\text{between segments}}, \tag{4}$$

where R_{kk} is the correlation matrix of segment k. The between-segment term is zero exactly when the segments move independently of one another: it is the systemic part. For two segments A and B, the mutual information $I(A;B)$ gives a contagion map in the same unit. Everything is unit-free: equity returns, rate changes and exchange rates can sit in the same matrix. We use the rank version of coupling.

OA.5.2 The euro-area panel

The panel is daily, from January 2005 to October 2026 (5,690 days), with 28 series: the overnight rate (EONIA, then €STR) for the money market; six ECB sovereign yields and spreads for bonds; five national indices for equities; nine banks and three insurers for intermediaries; four euro exchange

rates. The CISS uses the Euribor minus overnight spread for the money market; since Euribor is not available on the ECB portal, this segment for now contains only the overnight rate.

OA.5.3 Results

	Rise under	i.i.d. null		CCC-FHS null	
Component	stress (nats)	mean	p	mean	p
Total coupling	+1.61	+0.20	<0.001	+0.32	<0.001
Between segments	+0.19	+0.02	<0.001	+0.03	0.002
Within: equities	+0.46	+0.05	<0.001	+0.08	<0.001
Within: intermediaries	+0.48	+0.13	<0.001	+0.21	0.020
Within: bonds	+0.34	-0.01	<0.001	+0.00	<0.001
Within: FX	+0.14	+0.00	<0.001	+0.00	<0.001
Equities – intermediaries	+0.096	+0.010	<0.001	+0.016	0.012
Bonds – FX	+0.055	+0.004	<0.001	+0.005	0.002
Bonds – intermediaries	+0.040	+0.003	<0.001	+0.008	0.016
Bonds – equities	+0.029	+0.003	0.002	+0.004	0.013
Equities – FX	+0.046	+0.006	<0.001	+0.005	0.002
Intermediaries – FX	+0.040	+0.007	<0.001	+0.006	0.004
Money market – other segments	≈ 0	≈ 0	n.s.	≈ 0	n.s.

Table OA.6: Euro area: stress-minus-calm rise of each coupling component against the two placebo nulls (999 replications each; p exact, one-sided). Stress is the top decile of equity-block volatility, with the threshold computed on the past only. The CCC-FHS null fits a GJR-GARCH(1,1) to each of the 28 series.

Three readings (table OA.6 and figure OA.2).

- *Stress ties segments together*, far beyond both nulls: the between-segment rise is nine times the i.i.d. null's and six times that of the volatility-preserving null, which keeps each series' GARCH dynamics. Every pair outside the money market rises beyond both nulls; the links between equities and intermediaries and with FX are the clearest. The weakest result is coupling within the intermediaries block ($p = 0.020$ against the volatility-preserving null): bank and insurer stocks move together partly because their volatilities rise together.

- *The sovereign–bank link stands out.* Mutual information between bonds and intermediaries peaks in late 2011, at the height of the sovereign debt crisis (0.46 nat against 0.10 in late 2006). The aggregate CISS cannot show which segments an episode ties together.

- *The money market does not couple* in daily data: the overnight rate follows monetary policy, not market stress.

Against official indicators (table OA.7), total coupling flags recessions slightly better than the CISS and volatility, and it explains the CISS beyond what volatility explains. It does not, however,

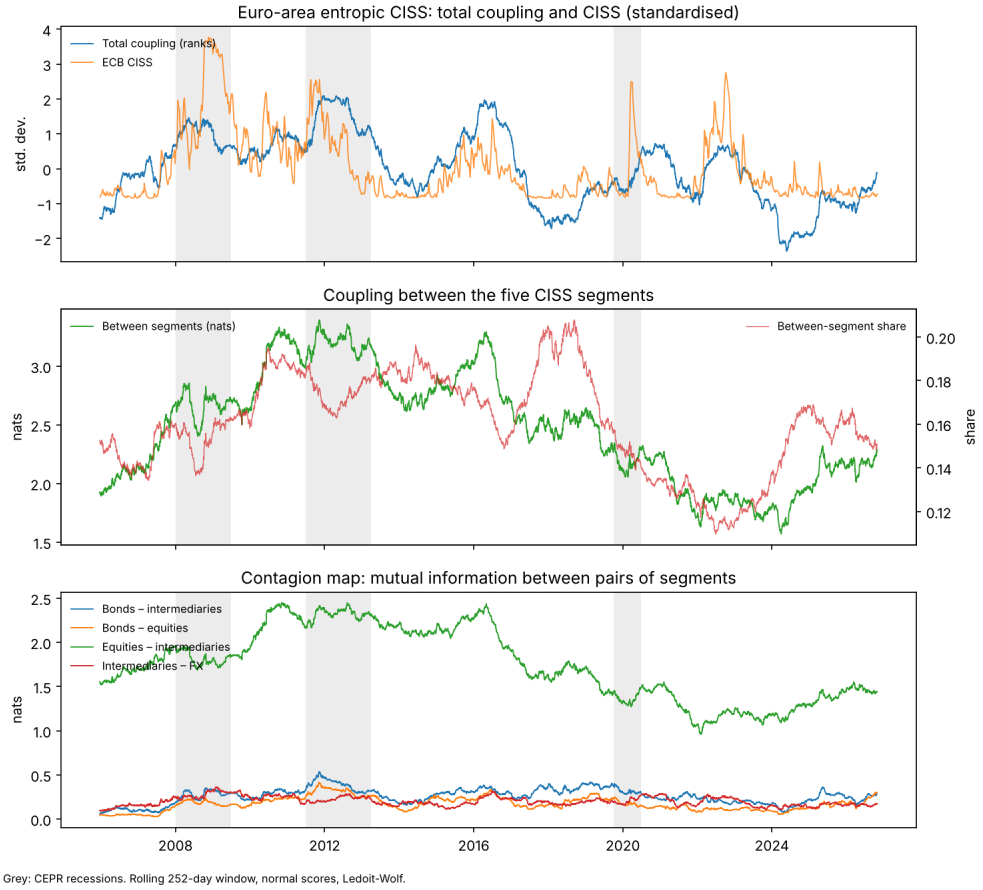


Figure OA.2: The euro-area entropic CISS. Top: total coupling and the ECB's CISS, standardised. Middle: coupling between the five segments, in nats, and its share of the total. Bottom: mutual information between pairs of segments. Grey areas: CEPR recessions.

	Total coupling	Between segments	Volatility	CISS
Correlation with the CISS	0.58	0.35	0.74	1
AUROC, CEPR recessions	**0.85**	0.75	0.80	0.83
Explains the CISS beyond volatility (t)	**6.7**	1.7		
Growth-at-Risk, alone: p	0.04	< 0.001	0.98	< 0.001
Growth-at-Risk, with CISS and past growth: p	0.78	0.81		< 0.001

Table OA.7: Euro area against official indicators. The AUROC is the probability that a randomly drawn recession date has a higher indicator than a non-recession date (0.5: no power; 1: perfect separation). Growth-at-Risk: regression of the bottom decile of four-quarter GDP growth on the indicator (Adrian et al., 2019), 79 quarters.

forecast growth once the CISS is accounted for. With three recessions and 79 quarters, these comparisons are indicative. The indicator is therefore *coincident*: a complement to the CISS that says which segments are linked, not a better predictor. On US data, against the St. Louis Fed stress index (Kliesen and Smith, 2010), coupling adds to volatility on the twenty stocks ($t \approx 4$ to 5), mainly before 2015, but not on the 48 industries.

Next steps. Add Euribor from its administrator (EMMI) to rebuild the money-market segment as in the CISS; date stress episodes with the ESRB crisis database (Lo Duca et al., 2017) and measure how the contagion map behaves around them; compare the indicator with institution-level systemic-risk measures, SRISK (Brownlees and Engle, 2017) and ΔCoVaR (Adrian and Brunnermeier, 2016).

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